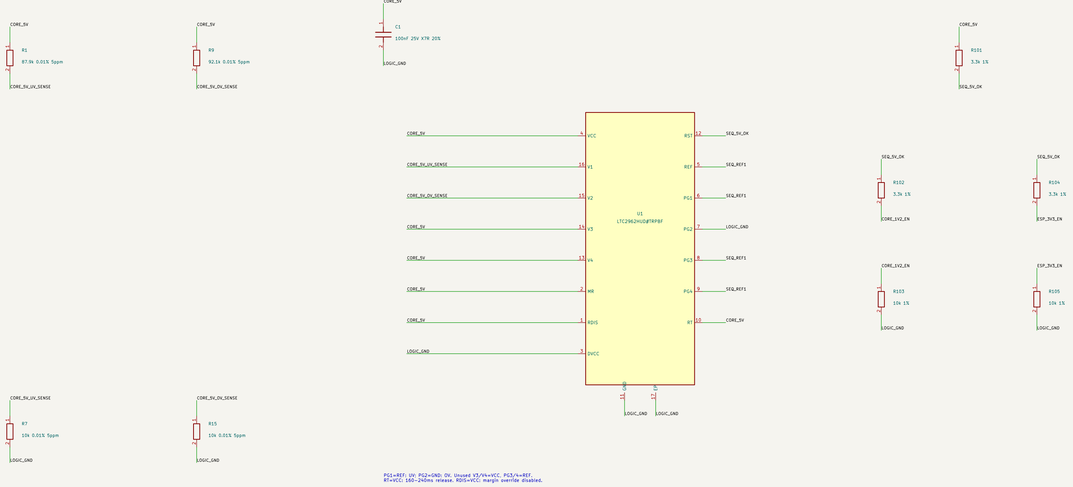
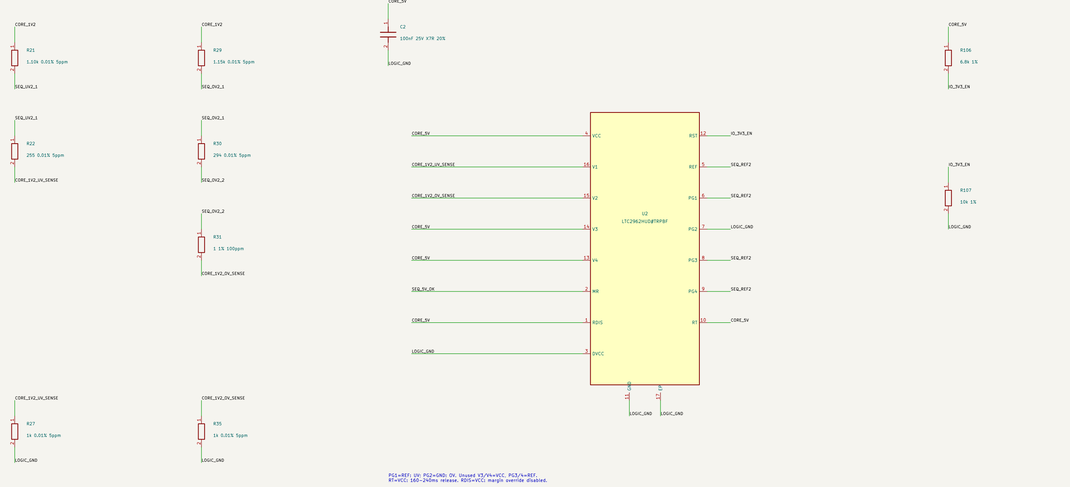
AUTONOMOUS POWER WINDOW / RESET CANDIDATE — NOT MACHINE PERMISSION

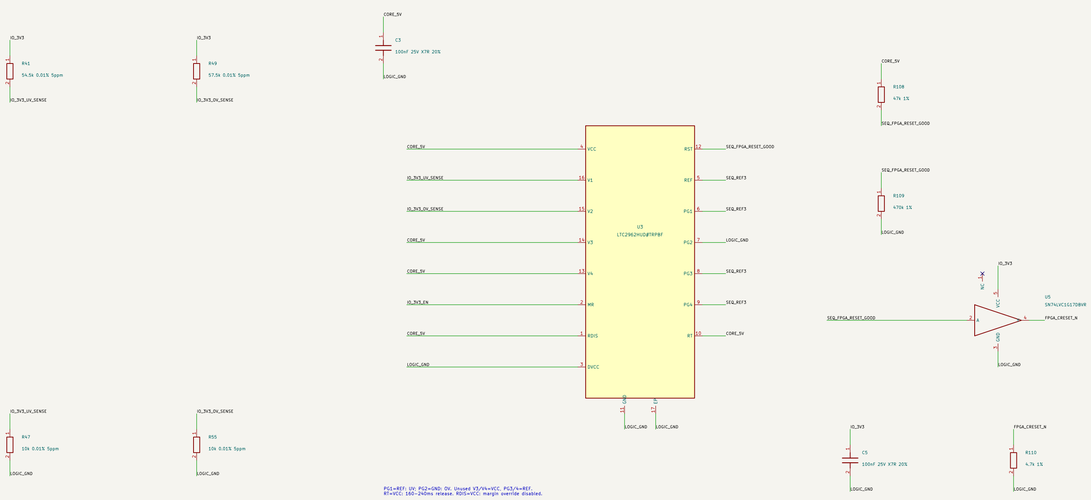
1. CORE_5V window / autonomous release



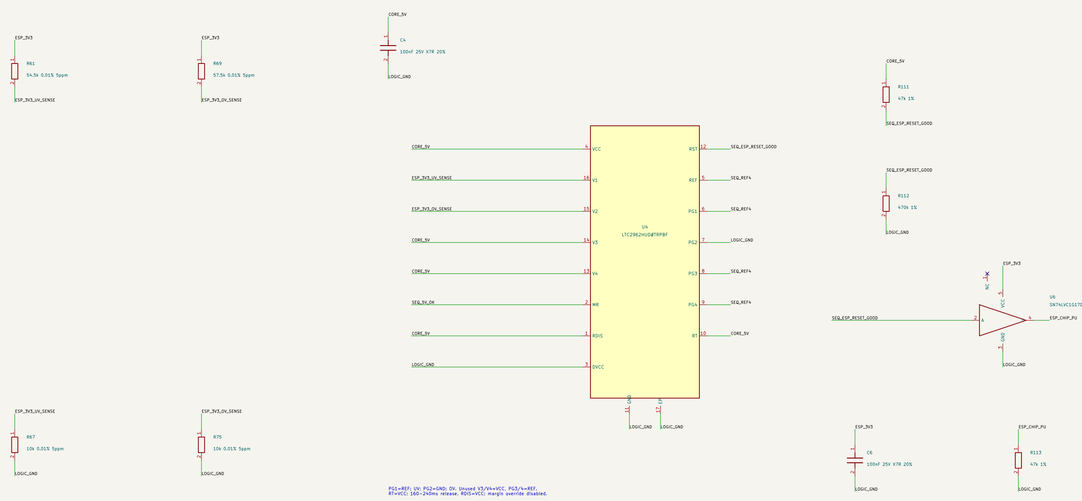
2. CORE_1V2 window / autonomous release

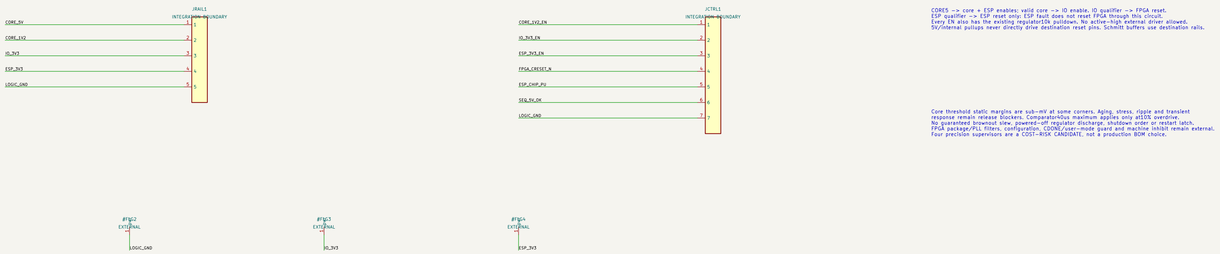


3. IO_3V3 window / autonomous release



4. ESP_3V3 window / autonomous release





CORES -> core + ESP enables; valid core -> IO enable, IO qualifier -> FPGA reset.
ESP qualifier -> ESP reset only: ESP fault does not reset FPGA through this circuit.
Every EN also has the existing regulator's EN pulldown. No active-high external driver allowed.
5V/internal pullups never directly drive destination reset pins. Schmitt buffers use destination rails.

Core threshold static margins are sub-mV at some corners. Aging, stress, ripple and transient response remain release blockers. Comparator+5us maximum applies only at10% overdrive.
No guaranteed brownout slew, powered-off regulator discharge, shutdown order or restart latch.
FPGA package/PLL filters, configuration, CDONE/user-mode guard and machine inhibit remain external.
Four precision supervisors are a COST-RISK CANDIDATE, not a production BOM choice.



Engineering candidate; no PCB or physical validation

Sheet: /
File: power_sequence.kicad_sch

Title: Autonomous rail qualification and reset

Size: A1	Date: 2026-09-28	**Rev: r0.1**
KiCad E.D.A. 10.0.4		Id: 1/1